A Systematic Comparison of Execution-Free Verification Formulations for Code Repair

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Abstract

LLM-based program repair increasingly generates multiple candidate patches, selecting the best via a verifier. Three execution-free alternatives to expensive execution-based verification have emerged: per-test binary classification (CLS), execution output generation (CWM), and trajectory-level patch scoring (TRAJ). No prior work compares them under matched conditions.

We present the first systematic comparison across two code LLM families (Qwen3, DeepSeek-Coder), two complexity levels, and multiple seeds under $\leq$8B LoRA fine-tuning. At function level, CLS and CWM are statistically equivalent across both families (TOST, 5 seeds): classification is not inherently easier than generation when outputs are short. TRAJ parity is not confirmed after multiple comparison correction at any scale tested, but the gap narrows monotonically from 3.11 pp (4B) to 1.07 pp (14B). At repository level, a significant interaction ($z = -23.7$, DeepSeek-Coder) confirms this equivalence collapses: the CLS–CWM gap widens to 18–26 pp across 2–3 seeds per model under our $\leq$8B LoRA protocol, driven by format collapse. At 14B (3 seeds), all three formulations narrow to within 1.07 pp, consistent with capacity-dependent resolution of formulation differences.

Error complementarity and a failure mode taxonomy motivate future work on complexity-aware formulation routing, though simple heuristics capture $<$0.4 pp of oracle advantage.

1 Introduction

LLM-based automated program repair systems increasingly rely on test-time compute scaling: generating K candidate patches and selecting the best via a verifier (Snell et al., 2025). Full execution verification is reliable but expensive ($1–3 per candidate at repository scale[1]), driving demand for execution-free alternatives. Three distinct formulations have emerged for this task: **generation-based simulation** (FAIR CodeGen team et al., 2025; Maimon et al., 2026) predicts execution output strings per test; **trajectory-level scoring** (Shum et al., 2025) predicts a binary patch verdict without per-test decomposition; and **per-test classification** (Ni et al., 2023; Chen et al., 2023) predicts a pass/fail token per test. Each makes different tradeoffs between diagnostic granularity, output complexity, and inference cost.

Despite active development along each axis, no prior work provides a systematic comparison holding model capacity, training data, and compute constant. Practitioners choosing a formulation for their verification pipeline must rely on results from different models, different datasets, and different evaluation protocols. Whether one formulation is uniformly preferable, or whether the choice depends on properties of the code being verified, remains an open empirical question. This connects to a broader pattern in NLP: task formulation—classification vs. generation vs. span extraction—systematically affects performance (Raffel et al., 2020), but the conditions under which each wins remain task-dependent and empirically unresolved. A related question is whether formulation choice matters at all relative to more basic decisions: at what complexity level does fine-tuning become necessary, and does formulation selection add value beyond what fine-tuning alone provides?

We present the first systematic empirical comparison across two representative code LLM families (Qwen3, DeepSeek-Coder), two complexity levels (function-level and repository-level), and multiple seeds. Our contribution is primarily methodological: a matched experimental framework that isolates formulation effects from confounding variables. The analysis also produces actionable outputs: a failure mode taxonomy that

[1] Based on per-instance costs reported by Yang et al. (2024) and Xia et al. (2025).

characterizes when and why each formulation fails (§3.5), error complementarity findings with practical ensemble implications (§3.6), and formulation selection recommendations conditioned on code complexity (§5). The results reveal a three-layer structure. Fine-tuning is necessary (0-shot→fine-tuned gains of 35–36 pp at repo level). After fine-tuning under our ≤8B LoRA protocol, *formulation choice is moderated by code complexity*: at function level, all formulations achieve comparable accuracy—consistent with the theoretical analysis in §2.2, which predicts that short outputs minimize generation-specific failures. At repository level under our ≤8B LoRA protocol, this equivalence collapses; variant-level analysis reveals three distinct failure modes consistent across model families, though gap magnitude depends on multiple confounds (§3.7). Figure 1 illustrates the three formulations.

We make the following contributions:

- **Complexity-moderated formulation choice.** We establish a three-layer structure: (1) fine-tuning is necessary (35–36 pp gains); (2) at function level, formulations are near-equivalent—CLS and CWM achieve statistical equivalence (TOST p=0.007, Holm-corrected) across both families, while TRAJ equivalence is not confirmed after Holm correction, but the gap narrows monotonically from 3.11 pp at 4B to 1.07 pp at 14B (§3.3); (3) at repository level under ≤8B LoRA, this equivalence collapses ($z = -23.7$, DeepSeek-Coder), with CWM falling 18–26 pp below CLS, driven by format collapse—four interventions all fail to close the gap. The TOST framework demonstrates strict equivalence rather than mere non-significance (§3.4, §3.5).
- **Error complementarity.** CLS and CWM exhibit partially complementary error patterns at function level (oracle: 90.63%, 95% CI [89.3%, 91.9%]; single-seed); at repo level, oracle routing reaches 95.83% variant accuracy (+14.58 pp; single-seed), the strongest practical argument for exploring formulation-level tradeoffs (§3.6).
- **Failure mode taxonomy.** Variant-level analysis reveals three distinct failure modes at repository level—format collapse (CWM), decision collapse (CLS under truncate preprocessing), and aggregation penalty (CLS under all-pass rule)—providing actionable diagnostic criteria for practitioners selecting a verification formulation (§3.5, Table 4).

2 Method

Let p denote a candidate patch and $T = \{t_1, \ldots, t_n\}$ its test suite, where each t_i has label $y_i \in \{\text{PASS}, \text{FAIL}\}$ from execution. Tests decompose into *fail-to-pass* (T_{f2p}) and *pass-to-pass* (T_{p2p}) subsets; a patch is correct iff $\forall\, t_i$: $y_i = \text{PASS}$. We compare three formulations for predicting correctness without execution.

2.1 Verification Formulations

Let c_p denote the code context (unified diff) of patch p.

CLS (per-test classification). $f(c_p, t_i) \to \hat{y}_i \in \{\text{PASS}, \text{FAIL}\}$ predicts a single binary token per test (cross-entropy loss). Patch verdict: $\hat{y}_p = \bigwedge_i [\hat{y}_i = \text{PASS}]$. Output cost: one token per test.

CWM (output generation). $g(c_p, t_i) \to \hat{o}_i \in \Sigma^*$ generates the execution output string (language modeling loss). Verdict by exact match: $\hat{y}_i = \text{PASS}$ iff $\hat{o}_i = o_i^*$.[2] When the generated output is unparseable (i.e., does not conform to the expected structured format), the prediction defaults to the majority class verdict. Output cost: $\sum_i |\hat{o}_i|$ tokens.

TRAJ (trajectory-level). $h(c_p, T) \to \hat{y}_p \in \{\text{RESOLVED}, \text{UNRESOLVED}\}$ predicts a single patch verdict from all tests and provides no per-test signal. TRAJ serves as a granularity control: CLS–TRAJ differences isolate the value of per-test decomposition independent of the classification formulation itself.

2.2 Expected Failure Modes

Each formulation has characteristic failure modes that differ in their sensitivity to code complexity:

Generation failure modes (CWM). The open-ended output space introduces three failure modes: (1) *token-budget exhaustion*, where $|o_i^*| > B$ guarantees exact-match failure due to truncation; (2) *string-value prediction errors*, where semantically equivalent outputs differing in surface form (floating-point precision, dictionary ordering) produce false negatives; and (3) *format compliance*

[2] Exact string matching is conservative, consistent with FAIR CodeGen team et al. (2025). Semantic equivalences are not counted.

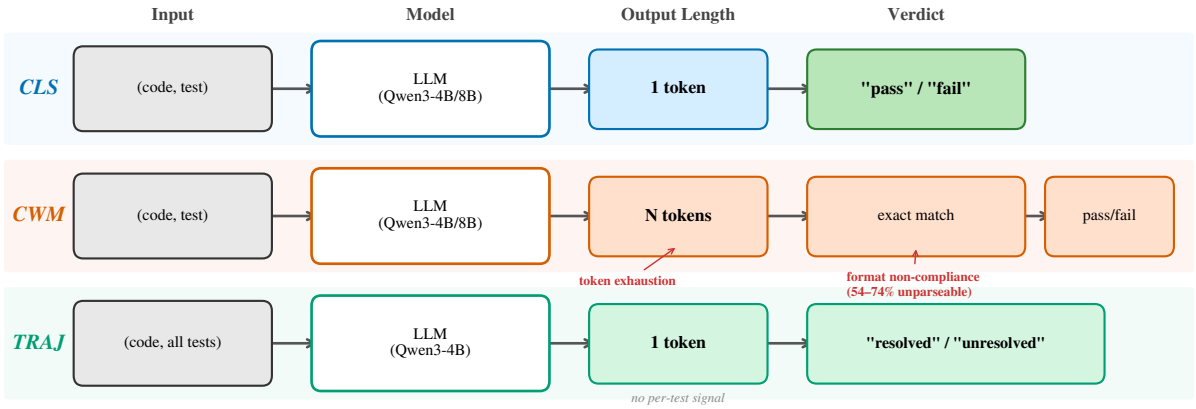


Figure 1: Three verification formulations. **Left:** CLS predicts binary pass/fail per test (1 output token). **Center:** CWM generates execution output, derives verdict by exact match ($O(L)$ tokens). **Right:** Trajectory-level scoring predicts patch-level resolved/unresolved (1 token, no per-test signal). Each formulation makes different tradeoffs between output complexity, diagnostic granularity, and inference cost.

collapse, where the model fails to produce structured predictions entirely at high input complexity, even when the generation budget is sufficient. Under optimistic independence, a token sequence of length L with per-token accuracy q has correct-sequence probability q^L; long outputs are therefore exponentially harder to predict exactly. We adopt exact string matching following the original CWM evaluation protocol (FAIR CodeGen team et al., 2025). Unparseable predictions default to the majority-class label (PASS).

Classification failure modes (CLS). The single-token output space eliminates format and truncation failures but constrains the model to encode its reasoning entirely within internal representations, without the intermediate reasoning trace that generation provides. CLS errors manifest as false positives or false negatives without format failures; at function level, a mild FP bias (56.5%) emerges, with the error direction depending on class distribution and fine-tuning dynamics.

Trajectory-level failure modes (TRAJ). By collapsing all per-test information into a single binary verdict, TRAJ discards diagnostic granularity. When a patch fails multiple tests, the trajectory-level formulation cannot identify *which* tests fail. This limits downstream utility (e.g., targeted repair) but avoids per-test aggregation errors.

Complexity dependence. These failure modes suggest a complexity-dependent interaction: when execution outputs are short and structured (function-level code), generation modes (1)–(2) are rarely triggered, and all formulations should perform comparably. As output length and structural complexity increase (repository-level code), generation-specific failures may compound, creating divergence. Whether this divergence reflects a fundamental property of the generation formulation or is contingent on model/tokenizer properties is an empirical question we investigate in §3.

2.3 Oracle Accuracy-Utility Framework

For a buggy patch with $k = |T_{\text{f2p}}|$ fail-to-pass tests, a per-test verifier with recall r detects the bug if it identifies at least one failing test: $P_{\text{detect}}(k) = 1 - (1 - r)^{k_{\text{eff}}}$, where $k_{\text{eff}} = k/(1 + (k - 1)\rho)$ accounts for within-patch correlation via ICC ρ (Shrout and Fleiss, 1979). The per-test advantage over a trajectory-level baseline (a_h) is $\Delta(k) = P_{\text{detect}}(k) - a_h$, with population expectation $\mathbb{E}[\Delta] = \sum_k P(|T_{\text{f2p}}| = k) \cdot \Delta(k)$. This predicts that per-test advantage concentrates in multi-F2P-test instances and vanishes at k=1, validated in §3 and Appendix A.

3 Experiments

We systematically compare CLS, CWM, and TRAJ across two representative code LLM families, two complexity levels, and multiple seeds to characterize the formulation landscape for neural code verification.

3.1 Experimental Setup

Datasets. *Function-level*: mutated variants from HumanEval (Chen et al., 2021) and MBPP (Austin et al., 2021) (five mutation types), labels verified by execution. 19,076 samples (train 15,290 / val 1,896 / test 1,890; pass:fail ≈ 52:48, majority baseline 51.75%). *Repo-level*: per-test instances from SWE-bench Verified (Chowdhury et al., 2024) mutants. ∼70K training samples; val (n=9,335) and test (n=9,784) have zero problem overlap (majority baseline 72.96%). Val-based checkpoint selection follows standard practice (Jimenez et al., 2024) where validation loss is computable; runs using truncate preprocessing produce NaN validation loss because repo-level execution traces often exceed the 2,048-token sequence length limit; after truncation, many samples have all target tokens removed (labels entirely −100), producing degenerate loss—a formulation-inherent weakness that classification and trajectory formulations avoid by design. NaN loss defaults to the final epoch checkpoint (affects DeepSeek CLS s123, DeepSeek CWM runs, and Qwen3 TRAJ runs; see Appendix E). Qwen3 CWM runs have valid validation loss with best checkpoints at ∼6% of training (step 300–400), indicating severe overfitting; s42 uses the trainer-selected best checkpoint, while s123/s456 use the final epoch checkpoint.

Models and training. We fine-tune Qwen3-4B and Qwen3-8B (Yang et al., 2025a) with Low-Rank Adaptation (LoRA) (Hu et al., 2022) (rank 16, 3 epochs) across five seeds. Seeds 42/123/456 use bfloat16; seeds 789/0 use NormalFloat 4-bit (NF4) quantization (Dettmers et al., 2023) to test robustness to precision reduction (function-level accuracy remains within seed noise across both precisions; see footnote in §3.4). As a cross-model consistency check, we additionally fine-tune DeepSeek-Coder-6.7B (Guo et al., 2024) with identical LoRA configuration (2 seeds each for CLS and CWM). Our CWM baseline is a LoRA fine-tuned variant for matched comparison, not a reproduction of the 32B Code World Model (FAIR CodeGen team et al., 2025). Repo-level models use val-selected checkpoints where available; runs with NaN validation loss use the final epoch checkpoint (Appendix E). Zero-shot baselines use Qwen3-8B and Qwen3-32B (NF4) as LLM-as-judge. Full hyperparameter details appear in Appendix E.

Table 1: Function-level per-test accuracy on HumanEval+MBPP (n=1,890). Mean ± std over seeds. Seeds 42/123/456: bfloat16; 789/0: NF4. DeepSeek: n=2 seeds; std reported for descriptive comparison, not inferential. TRAJ evaluated on 1,032-sample subset (447 matched variants). All prompting baselines use NF4 4-bit quantization for fair cross-scale comparison.

Method	Model	Seeds	Accuracy (%)
Qwen3 (primary)			
CLS (fine-tuned)	4B	5	**88.43 ± 0.78**
CWM (fine-tuned)	4B	5	87.19 ± 0.37
TRAJ (fine-tuned)	4B	5	86.30 ± 0.97
DeepSeek-Coder (cross-model consistency check)			
CLS (fine-tuned)	6.7B	2	**86.85 ± 0.11**
CWM (fine-tuned)	6.7B	2	86.83 ± 1.49
TRAJ (fine-tuned)	6.7B	2	84.16 ± 1.30
Prompting baselines			
0-shot	8B (Qwen3)	1	60.53
0-shot	14B (Qwen3)	1	66.24
0-shot	6.7B (DeepSeek)	1	63.00
ZS LLM-as-judge	8B (Qwen3)	1	64.40
ZS LLM-as-judge	6.7B (DeepSeek)	1	56.56
ZS LLM-as-judge	32B (NF4)	1	71.69
3-shot ICL	8B (Qwen3)	1	60.42
3-shot ICL	14B (Qwen3)	1	73.12
3-shot ICL	6.7B (DeepSeek)	1	59.95
Majority (pass)	—	—	51.75

Statistics. Confidence intervals use clustered bootstrap (Davison and Hinkley, 1997) (114 problem clusters, 10K resamples; we resample entire problems, not individual tests, to preserve within-problem correlations). Equivalence uses Two One-Sided Tests (TOST) (Schuirmann, 1987) with ±2 percentage point (pp) margin; for function-level TOST, we use seed-level paired analysis (df=4 for 5-seed comparisons, df=2 for 3-seed), treating each training seed as the unit of replication. This is more conservative than problem-level clustered bootstrap, which pools within-seed observations and may understate between-seed uncertainty. Holm-Bonferroni correction (Holm, 1979) controls family-wise error rate across the five primary comparisons (Table 15). Intra-class correlation coefficient (ICC) (Shrout and Fleiss, 1979) quantifies within-problem correlation. Variant-level effect sizes and statistical power analysis are reported in Appendix K. Oracle analysis uses SWE-bench Verified (500 instances).

3.2 Function-Level Results

Table 1 reports function-level results across formulations and model families.

All three formulations achieve comparable accuracy on Qwen3-4B: CLS 88.43% (±0.78), CWM 87.19% (±0.37), TRAJ 86.30% (±0.97).

Table 2: Model-size scaling: accuracy (%) at 4B, 8B, and 14B (Qwen3). Seeds 42/123 (4B/8B), seeds 42/123/456 (14B). All bfloat16 inference. Δ: change from previous size. CWM and TRAJ use n=2 seeds (4B/8B) and n=3 seeds (14B). TRAJ evaluated on 1,032-sample subset (447 matched variants); CLS/CWM use the full 1,890-sample test set.

Method	Size	s42	s123	s456	Mean	Δ
CLS	4B	87.83	89.52	—	**88.68**	—
	8B	**90.48**	89.40	—	**89.94**	+1.27
	14B	89.21	**90.79**	89.79	**89.93 ± 0.80**	−0.01
CWM	4B	87.51	87.20	—	87.36	—
	8B	87.67	86.98	—	87.33	−0.03
	14B	89.26	88.99	90.26	89.50 ± 0.67	+2.17
TRAJ	4B	85.66	85.47	—	85.57	—
	8B	87.60	87.79	—	87.70	+2.13
	14B	89.73	88.57	88.28	88.86 ± 0.77	+1.16

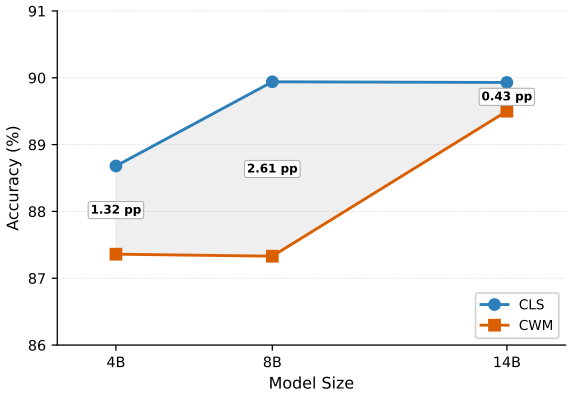


Figure 2: Function-level accuracy scaling (4B–14B). CLS and CWM accuracy with shaded gap region. The CLS–CWM gap follows an inverted-U trajectory: 1.32 pp at 4B, widening to 2.62 pp at 8B, then narrowing to 0.43 pp at 14B (3 seeds).

DeepSeek-Coder-6.7B shows: CLS 86.85%, CWM 86.83%, TRAJ 84.16%. The CLS–CWM gap (0.02–1.24 pp) is within seed noise in both families. However, the CLS–TRAJ gap is model-dependent: <1 pp on Qwen3 but 2.69 pp on DeepSeek-Coder, exceeding the ±2 pp equivalence margin (§3.4). CLS and CWM are robustly equivalent; TRAJ parity is model-contingent. Zero-shot baselines (Tables 1 and 3) establish that repo-level verification requires fine-tuning (0-shot→fine-tuned gains of 35–36 pp) and that formulation choice matters only at repo level (CLS outperforms CWM by 18–26 pp).

3.3 Scaling Behavior

Beyond overall accuracy, we examine whether formulation ranking is stable across model sizes. Table 2 compares all three formulations at 4B, 8B, and 14B. The 4B and 8B rows use seeds 42 and 123; the 14B row uses seeds 42, 123, and 456. All rows use bfloat16 inference.

CLS and TRAJ both scale positively from 4B to 8B (+1.27 pp and +2.13 pp respectively), while CWM remains flat (−0.03 pp). At 8B, the ranking is CLS (89.94) > TRAJ (87.70) > CWM (87.33). The CLS–CWM gap widens from 1.32 pp at 4B to 2.62 pp at 8B (Figure 2); clustered bootstrap on the 3-seed superset (42, 123, 456 for CLS/CWM) yields $p = 0.0066$ (95% CI $[1.80, 2.71]$), which confirms the widening is not a sampling artifact.

At 14B (3 seeds), all three formulations converge: CLS 89.93±0.80%, CWM 89.50±0.67%, TRAJ 88.86±0.77%. The maximum pairwise gap (CLS–TRAJ) narrows from 3.11 pp at 4B to 1.07 pp at 14B, within the ±2 pp equivalence margin. TOST confirms CLS–CWM equivalence (p=0.0006); CLS–TRAJ passes raw TOST (p=0.043, P(ROPE)=0.956) but not Holm correction (adj p=0.050; Appendix J.1). CWM's delayed scaling onset—flat from 4B to 8B, then +2.17 pp at 14B—and TRAJ's decelerating convergence (+2.13 pp at 8B, +1.16 pp at 14B) are both consistent with capacity-dependent convergence at function level.

Baseline scaling. Prompting baselines remain well below fine-tuning: the best result (14B 3-shot, 73.12%) is 16.82 pp below fine-tuned CLS at 8B. Only CLS scales positively across both complexity levels; TRAJ shows no repo-level scaling benefit (72.92% at both 4B and 8B).

3.4 Function-Level Equivalence

We test pairwise equivalence on 523 variant-level instances across five seeds.[3] Our ±2 pp SESOI threshold (Lakens, 2017; Wellek, 2010) is motivated by domain considerations but was not pre-registered: at 85–88% accuracy, 2 pp corresponds to ∼2–3 problems out of 114, below practitioner-relevant differences, and is consistent with between-seed variation (0.37–0.97 pp std). Within-problem predictions are correlated (ICC = 0.191, DEFF = 3.97 (Moulton, 1990)); confidence intervals use clustered bootstrap (§3.1). Sensitivity analysis confirms equivalence holds at

[3] Seeds 789/0 use NF4 inference; the remaining three use bfloat16. Restricting to bfloat16 only: CLS–CWM remains significant (p<0.05); CLS–TRAJ yields p=0.128.

margins as tight as ±1.52 pp; conclusions are sensitive to margin choice (Table 13).

CLS–CWM equivalence. The pooled gap is 1.24 pp (88.43%–87.19%); TOST confirms equivalence at $p = 0.007$ (±2 pp margin), surviving Holm-Bonferroni correction (k=5, adjusted p=0.028; Table 15). Equivalence holds at margins as tight as ±1.52 pp (Table 13). On DeepSeek-Coder, the CLS–CWM gap of 0.02 pp is consistent with equivalence but underpowered (2 seeds).

CLS–TRAJ equivalence. The pooled gap is −0.45 pp (95% CI [−1.71, +0.84]); TOST yields $p = 0.025$ (±2 pp), nominally significant but not surviving Holm-Bonferroni correction (k=5, adjusted p=0.075; Table 15). The DeepSeek CLS–TRAJ gap (2.69 pp) exceeds the ±2 pp margin; TRAJ parity is model-dependent. Per-test classification provides diagnostic information (which tests fail) at no cost to variant-level accuracy, paralleling process supervision (Lightman et al., 2023).

Statistical power. With $n_{\text{eff}} \approx 86$ after clustering, TOST power is ∼0.3 (Cohen, 1988)—a 70% probability of failing to confirm equivalence even if the true difference is moderate. The CLS–CWM equivalence (p=0.007) is robust to this limitation; the CLS–TRAJ non-confirmation (p=0.025, adjusted p=0.075) may reflect insufficient power rather than a genuine difference (a minimum $n_{\text{eff}} \approx 350$ would be needed for ≥0.8 power at ±2 pp). A Bayesian ROPE analysis corroborates: $P(\text{ROPE})$=0.975 for CLS–CWM and $P(\text{ROPE})$=0.970 at variant level for CLS–TRAJ (Appendix J.1). At per-test level, CLS–TRAJ $P(\text{ROPE})$=0.542 (inconclusive), reflecting a larger per-test gap (+1.93 pp) that shrinks at variant level (−0.45 pp) through aggregation effects. 14B scaling data independently addresses this limitation: the CLS–TRAJ gap narrows monotonically from 3.11 pp to 1.07 pp with raw TOST confirming equivalence at 14B (p=0.043, $P(\text{ROPE})$=0.956; §3.3), consistent with the equivalence interpretation.

3.5 Repo-Level: Complexity-Dependent Divergence

Table 3 compares formulations at repository-level complexity. Variant-level comparisons (n=48) are exploratory due to limited seed coverage; the per-test interaction effect is statistically confirmed (see below). Checkpoints are val-selected where validation loss is computable (§3.1); the test set has zero problem overlap with validation.

Table 3: Repo-level cross-formulation comparison on SWE-bench Verified test set. Qwen3-8B CLS/TRAJ: mean ± std over 3 seeds; DeepSeek CLS: 2 seeds‡; DeepSeek TRAJ: 3 seeds; Qwen3-4B: seed 42. CWM: 3 seeds (42/123/456).

Model	CLS (%)	CLS Var. (%)	CWM (%)	TRAJ Var. (%)◇
Qwen3-4B	83.63	64.58	58.04	72.92
Qwen3-8B	**84.83 ± 0.67**	72.92 / 81.25†	66.48 ± 1.15	72.91 ± 3.61
DeepSeek-6.7B	84.38 ± 2.50‡	81.25†	58.30 ± 2.82	77.08 ± 3.61
0-shot (Qwen3-8B)	49.90	—	—	—
0-shot (DeepSeek)	48.54	—	—	—
Majority	72.96	54.17	72.96	54.17

†Individual-seed variant accuracy. ‡One seed excluded: truncate preprocessing→NaN val loss, decision collapse (Table 4); mean over two matched seeds. ◇Var. columns: variant-level accuracy (n=48); CLS aggregated via all-pass rule; TRAJ by design. Per-test: n=9,784. CWM unparseable: 54% (4B); default to pass; parseable-only 38–43%, below majority. Eval seq_length differs across models (Appendix O). CWM uses final-epoch checkpoint (step 6633) for s123/s456; s42 uses an earlier checkpoint (step 400) from a prior training configuration. Val loss is undefined for DeepSeek CWM runs (Qwen3 CWM has valid val loss; see §3.1). Checkpoint sensitivity analysis in Appendix O.

Few-shot prompting (3-shot ICL) collapses to an always-predict-unresolved baseline (variant accuracy 45.83%, below majority 54.17%; Appendix B), establishing that execution-free verification requires task-specific fine-tuning regardless of formulation. The fine-tuning requirement (35–36 pp gains, Table 3) is the first layer of our three-layer structure.

Formulation divergence. Under our 4–8B LoRA protocol, CLS and CWM diverge: the gap is 25.59 pp at 4B and 18.35 pp at 8B (Table 3), with CWM falling below majority baseline at both scales. Limited seed coverage and multiple confounds (§3.7) limit causal attribution.

Cross-model consistency. DeepSeek-Coder shows the same direction (CLS > CWM) across all seeds; gap variation across families (20 pp Qwen3 vs. 23–28 pp DeepSeek) narrows to ∼2 pp under matched checkpoints (Appendix O). All six CWM repo-level seeds exhibit format collapse (>50% unparseable), ruling out seed-specific artifacts.

Failure mode taxonomy. Variant-level analysis (n=48) reveals three failure modes (Table 4): format collapse (CWM, 51–74% unparseable), decision collapse (CLS under truncate preprocessing, fail recall ∼14%), and aggregation penalty (CLS

Table 4: Repo-level failure mode taxonomy. Variant-level majority baseline: 54.17%.

Failure Mode	Representative	Fail Recall	Variant Acc
Format collapse	CWM (all)	N/A	48–52%
Decision collapse	CLS (DeepSeek, s123)	∼14%	56%
Aggr. penalty	CLS (Qwen3-8B)	51–57%	73–81%
Aggr. penalty	CLS (DeepSeek, s42)	38–51%	79–81%

under drop preprocessing, fail recall 38–57%). Decision collapse is configuration-dependent, not model-inherent. TRAJ avoids aggregation penalty by predicting variant outcomes directly (72.91–77.08%). Downsampling CLS to match TRAJ's data volume still yields 80.74% (+7.82 pp above TRAJ; Appendix B). The CLS–TRAJ repo-level comparison remains exploratory: downsample ablation controls data volume but not label granularity (per-test vs. variant-level), a formulation-inherent confound that cannot be experimentally resolved.

Default assignment artifact. The apparent CWM accuracy increase from 4B to 8B is an artifact of rising unparseable rate (53.7%→73.8%), inflating accuracy via majority-class default while parseable-only accuracy decreases (42.88%→38.52%); cross-seed evidence confirms the pattern (Appendix N).

Complexity as moderator. The CLS–CWM gap widens from 1–3 pp (function) to 18–26 pp (repo) under our protocol; a matched-context ablation confirms input disparity explains <3% of this gap (Appendix B).

Formulation×complexity interaction. A mixed-effects LPM (n=56,480, 139 clusters, DeepSeek-Coder) confirms: CWM effect at function level is negligible ($\hat{\beta}=-0.002$, p=0.805), but the formulation×complexity interaction is highly significant ($z=-23.7$, $p<0.001$). A cross-model LPM confirms formulation dominates ($\hat{\beta}=-0.22$ vs. model $\hat{\beta}=+0.011$); model identity significance is estimator-dependent (LPM p=0.002; GEE p=0.449), consistent with limited clusters (k=25 < 40; Appendix M).

3.6 Analysis

Aggregation penalty. The aggregation penalty ranges from 1.36 pp to 19 pp, negligible at function level ($\bar{T} \approx 2.3$) but substantial at repo level ($\bar{T} \gg 10$; Appendix I(a)).

Why does generation degrade at repo level? Output complexity increases nonlinearly: function-level outputs average ∼5–15 tokens; repo-level outputs average ∼477 tokens, causing format collapse (53–74% unparseable). Even parseable outputs achieve only 38–43%, below majority; forced parsing confirms the bottleneck is semantic, not syntactic (Appendix I(c)). Four targeted interventions all fail to close this gap: doubling LoRA rank to 32 does not improve format compliance; majority voting (k=5) is ineffective because collapsed outputs are near-identical across samples; relaxed matching still leaves >50% unparseable; and scaling from 4B to 8B worsens the unparseable rate from 53.7% to 73.8% (Appendix I(g)). Checkpoint selection does not explain the gap: matched epoch-3 comparisons yield 20–23 pp gaps, and CLS varies <1 pp across epochs (Appendix O). CLS achieves 217× higher throughput than CWM (1 vs. avg 477 output tokens; Appendix I(i)).

Error complementarity. CLS and CWM exhibit partially complementary errors. An oracle combination achieves 90.63% (+2.80 pp; 95% CI [89.3%, 91.9%]; single-seed) at function level and 95.83% (+14.58 pp; n=48) at repo level. Simple routing heuristics capture <0.4 pp of the oracle gap; the complementarity is instance-specific, suggesting a learned router is needed (Appendix A).

3.7 Threats to Validity

We identify five potential confounds and assess each in Appendix O: checkpoint selection (<1 pp variation in ablation; NaN validation loss forces final-epoch default in DeepSeek CWM runs and DeepSeek CLS s123), evaluation sequence length (∼1.5 pp), preprocessing (∼8 pp cross-configuration variance), TRAJ training regime (downsample confirms formulation advantage), and CWM scale/pretraining (primary limitation; all repo-level CWM conclusions conditional on the ≤8B LoRA regime). Preprocessing and checkpoint interact: truncate preprocessing produces NaN validation loss, forcing final-epoch checkpoint selection; this chain accounts for up to 7.68 pp accuracy variation within the same model and formulation (DeepSeek CLS s42: 82.61% vs. s123 truncate: 74.93%; Appendix O). Repo-level paired comparisons use different random seeds across formulations; while function-level seed noise is small (<2 pp), repo-level vari-

ance can be larger (DeepSeek CLS std = 2.50 pp across matched seeds), limiting paired inference. The 18–26 pp gap exceeds any single assessed confound, but correlated confounds and limited seeds mean definitive causal attribution requires multi-seed replication. Holm-Bonferroni correction (k=5) confirms the two core claims survive (adjusted p=0.015 and p=0.028; Table 15): repo-level CLS–CWM divergence and function-level CLS–CWM equivalence.

4 Related Work

Execution Simulation and Verification. Execution trace generation improves code reasoning (Nye et al., 2021), with extensions to output prediction (FAIR CodeGen team et al., 2025; Maimon et al., 2026; Liu et al., 2024, 2023; Lee and Huang, 2026), failure analysis (Abdollahi et al., 2025), and error detection (Li et al., 2024). CRUXEval (Gu et al., 2024) benchmarks code execution prediction at function level; RepoExec (Nam et al., 2025) evaluates repo-level code generation with execution-based correctness checks. Our work extends execution prediction to repo-level repair verification. CodeScaler (Zhu et al., 2026) trains execution-free reward models for both RL and test-time scaling. Verification formulations span trajectory-level reward (Shum et al., 2025), process-level feedback (Li et al., 2025b; Xu et al., 2025), per-test classification (Ni et al., 2023; Chen et al., 2023), learned metrics (Zhuo, 2024; Dong et al., 2024; Zhou et al., 2023), and RL with execution-based rewards (Wang et al., 2026; Tang et al., 2026). Aletheia (Venkatkrishna et al., 2026) compares training recipes; we compare *formulations*. Test-time scaling (Snell et al., 2025; Li et al., 2025a; Chen et al., 2025) and self-consistency (Wang et al., 2023) demonstrate verification value; none compare formulations at matched capacity.

Process vs. Outcome Supervision. PRMs outperform ORMs for math reasoning (Lightman et al., 2023; Uesato et al., 2022), with automated annotation (Wang et al., 2024; Luo et al., 2024) and code-specific extensions (Zhang et al., 2025; Ye et al., 2025; Zhang et al., 2026; Jiang et al., 2025). GenRM (Hosseini et al., 2025) and GenPRM (Zhao et al., 2026) show that generative verifiers outperform discriminative ones for math reasoning; our results reveal a domain difference—in code verification, the generative formulation (CWM) does not outperform the discriminative one (CLS), suggesting that the generation advantage may be domain-specific. CodeRM (Ma et al., 2025) demonstrates that test-level reward granularity improves code generation quality, corroborating our finding that per-test classification provides diagnostic value. In reward modeling terminology, CLS corresponds to process-level (PRM-like) verification while TRAJ provides outcome-level (ORM-like) feedback; CWM serves as a generative verifier (Hosseini et al., 2025). Our CLS provides per-test (process) supervision while TRAJ provides outcome-level; the PRM literature compares supervision granularity given a fixed output format, while we additionally vary the output format (CWM generation vs. CLS classification).

Repair, Mutation Testing, and Task Framing. CodeRL (Le et al., 2022), AlphaCode (Li et al., 2022; AlphaCode Team, 2023), and Reflexion (Shinn et al., 2023) use execution feedback for repair. SWE-bench (Jimenez et al., 2024; Chowdhury et al., 2024) provides benchmarks; SWE-agent (Yang et al., 2024) and Agentless (Xia et al., 2025) demonstrate pipelines where verification is a bottleneck (Yang et al., 2025b). Predictive mutation testing (Tufano et al., 2019; Kim et al., 2022; Wang et al., 2025; Jain et al., 2023) shares our per-test structure. Raffel et al. (2020) showed that task framing affects performance across NLP tasks; we extend this to code verification, finding that complexity moderates the classification–generation tradeoff.

5 Conclusion

We presented the first systematic comparison of three execution-free verification formulations (CLS, CWM, TRAJ). At function level, CLS and CWM are statistically equivalent (TOST p=0.007, 5 seeds); TRAJ equivalence is not confirmed after correction, but the gap narrows monotonically to 1.07 pp at 14B. Complementary errors (oracle: 90.63%/95.83%; single-seed) motivate complexity-aware routing. At repository level under ≤8B LoRA, divergence reaches 18–26 pp ($z = -23.7$), with CWM falling below majority baseline—consistent across families but dependent on multiple confounds (§3.7). 14B results (3 seeds) corroborate function-level convergence: the CLS–CWM gap narrows to 0.43 pp (TOST p=0.0006) and the CLS–TRAJ gap to

1.07 pp, consistent with capacity-dependent scaling; whether this extends to repo level remains open. This contrasts with mathematical reasoning, where generative verification outperforms discriminative (Hosseini et al., 2025), suggesting the generation advantage is domain-dependent rather than universal.

Three directions follow: (1) complexity-aware routing; (2) capacity scaling at 32B+ (FAIR CodeGen team et al., 2025); (3) per-test diagnostics for downstream repair.

Limitations

Primary findings use $\leq$8B LoRA fine-tuning with multi-seed replication; 14B results use a single model family (Qwen3) with 3 seeds; cross-family replication and evaluation at 32B+ are needed. The repo-level gap may diminish at 32B+ scale or with execution-trace pretraining (FAIR CodeGen team et al., 2025). All evaluation uses Python-only benchmarks; other languages may exhibit different format compliance rates. Training data uses synthetic mutation operators (Andrews et al., 2005; Just et al., 2014); the synthetic-to-real transfer gap remains untested (Appendix O).

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Appendix overview. A: Oracle resolve advantage. B: Ablation studies. C-F: Extended analysis (error complementarity, parseable accuracy, forced parsing, qualitative errors, failed interventions, mutation scaling, inference efficiency, re-repair, chain-of-thought). G: TOST sensitivity (G.1: Bayesian ROPE). H: Effect sizes. I: Holm-Bonferroni. J: Mixed-effects regression. K: Parseable bias. L: CWM convergence. M: Few-shot. N: Threats. O: Downsample. P: Limitations.

A Oracle Accuracy-Utility Analysis

We stratify SWE-bench Verified instances by F2P test count T, comparing per-test vs. trajectory-level oracle resolve rates under i.i.d. and block-correlated error models (ICC $=$ 0.191, from function-level analysis; §3.4).

Table 5: Oracle accuracy-utility frontier: per-test advantage stratified by F2P test count (T) on SWE-bench Verified (500 instances). Block-correlated model uses function-level ICC $=$ 0.191 (from function-level analysis, §3.4). P2P regression rate: 1.04% mean, 0% median.

T	Inst.%	i.i.d. (pp)	Block (pp)	Contrib.%
1	68.7	−0.04	−0.09	≈0
2	19.1	18.49	31.69	55.3
3–5	7.1	25.00	40.74	≈26
≥ 6	5.1	30.00	40.06	≈19
All	100.0	—	**10.93**	100.0

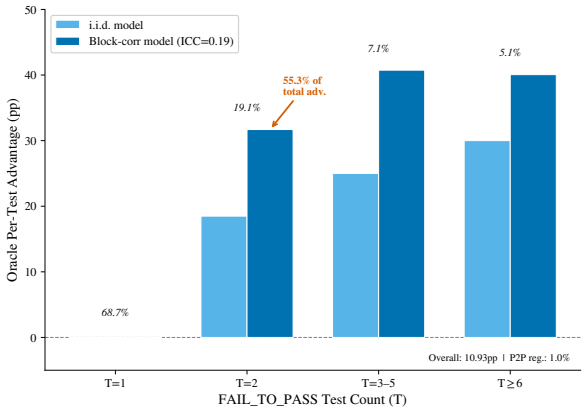


Figure 3: Oracle per-test advantage by F2P test count T. Advantage is zero at T=1 (68.7%) and concentrates at T=2 (19.1%, contributing 55.3% of total advantage). Note: y-axis does not start at zero.

Table 5 and Figure 3 reveal a sharp concentration: at $T = 1$ (68.7% of instances), per-test and trajectory-level are informationally equivalent. The advantage materializes at $T = 2$ (19.1%), where block-correlated advantage reaches 31.69 pp (55.3% of total). The overall block-correlated advantage is 10.93 pp, driven by the $T \geq 2$ minority. Block correlation amplifies relative to i.i.d.: at $T = 2$, 31.69 pp vs. 18.49 pp (1.7×). P2P regression is negligible (1.04% mean, 0% median); the practical value concentrates in F2P discrimination.

Sensitivity to ICC estimate. The analysis above uses ICC $=$ 0.191 from function-level analysis (§3.4), where mean cluster size $\bar{k} \approx 16.5$ yields DEFF $= 3.97$. At repository level, within-problem correlation is substantially stronger: ICC $= 0.3146$ with DEFF $= 114.4$ (§3.5), yielding an effective sample size of $n_{\text{eff}} \approx 9{,}784/114.4 \approx 86$ independent observations (Moulton, 1990)—still sufficient for regression but limiting precision of variance component estimates. This is a consequence of the larger and more heterogeneous test suites in SWE-bench Verified problems. Under higher ICC, within-problem errors cluster more tightly and amplify the block-correlated per-test advantage at each T stratum; the correction from i.i.d. to block-correlated grows monotonically with ICC. The function-level ICC $=$ 0.191 used in Table 5 thus provides a conservative lower bound on the per-test advantage. The repo-level correlation structure would yield an even larger gap between per-test and trajectory-level oracle resolve rates, which strengthens the case for per-test

granularity at repository-level complexity.

B Ablation Studies

LoRA rank sensitivity. All CWM experiments use LoRA rank r=16. To test whether this capacity constraint handicaps CWM, we train rank-32 variants (seed 42), doubling adapter parameters.

Table 6: LoRA rank ablation for CWM 4B (seed 42, 3 epochs, lr $= 2\times10^{-4}$). Rank-16 seed variance: 0.53 pp (seeds 42 vs. 456).

Config	α/r	Acc (%)	Fail F_1	Pass F_1	Δ
r=16, α=32	2	**87.51**	—	—	—
r=32, α=32	1	87.00	87.1	86.9	−0.51
r=32, α=64	2	87.04	87.27	86.79	−0.47

Both rank-32 variants fall within the 2-seed variance of the rank-16 baseline (0.53 pp between seeds 42 and 456; Table 6). The α=64 variant (matching the baseline α/r=2 ratio) rules out the confound that the α=32 result might reflect reduced effective scaling rather than rank insensitivity. This is consistent with CWM's function-level performance being bounded by output-format constraints rather than adapter capacity.

Majority voting. Majority voting (K=3) decreases accuracy from 87.51% to 86.83% (−0.68 pp): errors are systematic rather than stochastic, so sampling-based ensembles do not help.

Sequence-length ablation details. Both formulations are trained with `max_seq_length`=2,048. At inference, CWM reserves `max_new_tokens`=512, leaving ∼1,536 effective input tokens; CLS allocates nearly all context to input (5,120–8,192 tokens). A matched-context ablation (CLS 8B at 2,048 tokens) achieves 83.04%, only 1.44 pp below the 8,192-token evaluation (84.48%), though truncation introduces 136 unparseable outputs (1.39% of test samples). The matched-context gap (CLS@2,048 vs. CWM@2,048) is 25.00 pp (98% of the full 25.59 pp gap), confirming that input context disparity explains at most 3% of the difference.

Training data size control. To control for the training set size asymmetry between CLS (∼70K instances) and TRAJ (386 variant-level instances, ∼8K per-test equivalents, at repo level), we downsample CLS training data to match TRAJ in data volume. Downsampled CLS (Qwen3-4B, seed 42) achieves 80.74% repo-level per-test accuracy, only 2.89 pp below the full-data CLS (83.63%) and still 7.82 pp above TRAJ (72.92%). The CLS formulation advantage is not attributable to training data volume.

Few-shot prompting baselines. We evaluate 3-shot in-context learning at repo level to test whether prompting strategy closes the gap to fine-tuned models. Table 7 reports both per-test and variant-level accuracy.

Table 7: Repo-level prompting baselines (per-test n=9,784; variant-level n=48). Majority baseline: 72.96% per-test, 54.17% variant-level.

Model	0-shot (per-test)	3-shot (per-test)	3-shot (variant)
Qwen3-8B	49.90%	71.61%	45.83%*
DeepSeek-6.7B	48.54%	46.28%	45.83%*

*Both models achieve 45.83% by correctly classifying all 22 unresolved variants and zero resolved variants (22/48 = 45.83%), equivalent to an always-predict-unresolved baseline. The 100% overlap in variant-level predictions confirms complete degradation at task-relevant granularity.

Both models converge to exactly 45.83% variant-level accuracy under 3-shot prompting, below majority baseline (54.17%). Variant-level analysis reveals this is not a coincidence: both models correctly classify all 22 unresolved variants and zero resolved variants, equivalent to an always-predict-unresolved baseline with 100% prediction overlap. Per-test accuracy diverges sharply (Qwen3-8B 71.61% vs. DeepSeek 46.28%), but this divergence is entirely superficial: Qwen3-8B predicts pass on nearly all tests, inflating per-test accuracy on a pass-heavy dataset (72.96% pass rate) without genuine discriminative ability. Per-test metrics can thus severely misrepresent baseline capability; variant-level evaluation is necessary to detect complete degradation. Neither 0-shot nor few-shot prompting approaches fine-tuned performance (82–84% per-test); fine-tuning is a necessary condition for repo-level verification.

C Repo-Level Per-Mutation Breakdown

Table 8 compares per-mutation accuracy at repo level. CLS produces 0% unparseable outputs; all values are direct classification accuracy. CWM overall accuracy includes default-pass assignment for unparseable outputs (see §3.5).

Table 9 shows that CWM generation quality degrades from 4B to 8B across all mutation types.

Table 8: Per-mutation test accuracy (%) at repo level (SWE-bench Verified, n=9,784). CLS 8B reports two seeds; Δ is mean 4B→8B gain for CLS. CWM 4B per-mutation overall accuracy omitted for non-gold mutations due to low parse rates (4.4–56.9%, Table 9); gold overall accuracy derived from parse rate 43.5% and parseable accuracy 23.5%.

Mutation	n	CLS 4B	CLS 8B s42	CLS 8B s123	CWM 4B	CWM 8B	Δ CLS
gold	4,633	98.25	96.24	99.60	66.72	80.06	−0.33
operator_swap	3,357	78.91	81.38	82.60	—	54.57	+3.08
off_by_one	183	58.47	86.89	79.20	—	63.39	**+24.58**
return_value	562	67.97	75.44	66.40	—	49.11	+2.95
boundary_removal	1,049	46.90	46.90	44.90	—	42.23	−1.00

Table 9: CWM parse rates and parseable-only accuracy per mutation type (seed 42). Parse rate decreases uniformly from 4B to 8B; parseable-only accuracy is below random binary chance (50%) for gold at both scales.

	Parse Rate (%)		Parse Acc (%)	
Mutation	**4B**	**8B**	**4B**	**8B**
gold	43.5	23.9	23.5	16.5
operator_swap	56.9	21.2	63.1	58.8
off_by_one	4.4	27.3	0.0	38.0
return_value	48.8	26.9	27.0	56.9
boundary_removal	30.6	51.8	59.2	51.8

Key findings: (1) Gold parse rate collapses from 43.5% to 23.9% at 8B; the simplest generation target (unmodified code, where all tests pass) degrades most, consistent with a fundamental generation bottleneck rather than task difficulty. (2) Among parseable outputs, gold accuracy is 16.5–23.5%: the model generates outputs that fail exact-match even for correct code. (3) `off_by_one` is an anomaly: parse rate *increases* from 4.4% to 27.3%, but this is from an extremely low base (n=8 parseable at 4B), making both directions unreliable for this category.

D Variant-Level Accuracy

Table 10 reports variant-level accuracy (n=48) alongside per-test accuracy for all repo-level configurations. For CLS, the aggregation penalty (per-test → variant-level drop) ranges from 1 pp to 19 pp and is largest for models with low fail recall under the all-pass rule. CWM variant-level accuracy (48–52%) falls at or below the majority baseline, consistent with format collapse as the dominant failure mode.

E Training Hyperparameters

Table 11 lists all hyperparameters used for fine-tuning. Both CLS and CWM share the same LoRA configuration, optimizer, and training schedule; the only differences are per-device batch size, gradient accumulation steps (adjusted to maintain identical effective batch size of 32 across model sizes), and learning rate (DeepSeek-Coder uses 2×10^{-4} vs. 5×10^{-5} for Qwen3, selected via grid search over $\{5\times10^{-5}, 1\times10^{-4}, 2\times10^{-4}\}$ on held-out validation loss).

Table 10: Repo-level per-test vs. variant-level accuracy (n=48 variants). Penalty = per-test − variant-level. TRAJ predicts variant outcomes directly (no aggregation). Variant-level majority baseline: 54.17% (26/48).

Model	Seed	Per-test (%)	Variant (%)	Penalty (pp)	Fail Recall (%)
CLS					
Qwen3-8B	s42	84.48	72.92	11.56	56.76
Qwen3-8B	s123	85.60	81.25	4.35	51.44
Qwen3-4B	s42	83.63	64.58	19.05	48.72
DeepSeek-6.7B	s42 (e1)	82.61	81.25	1.36	51.02
DeepSeek-6.7B	s42 (e3)	82.51	79.17	3.34	38.47
DeepSeek-6.7B	s123	74.93	56.25	18.68	∼13.72
CWM					
Qwen3-8B	s42	65.17	52.08	13.09	—
DeepSeek-6.7B†	s123	—	47.92	—	—
TRAJ					
Qwen3-4B	s42	—	72.92	—	—
Qwen3-8B	s42	—	70.83	—	—
DeepSeek-6.7B	3-seed	—	77.08 ± 3.61	—	—
Majority	—	72.96	54.17	—	—

†Evaluated on ∼75% of test instances (7210–7668 of 9784). DeepSeek TRAJ Table 3 reports 3 seeds (42/123/456) for alignment with CLS/CWM seed coverage; three supplementary seeds (0/789/999) yield 75.0%, 66.7%, 79.2% respectively (6-seed mean 75.35 ± 5.00%), consistent in direction.

Data processing. DeepSeek CWM seeds use `train_unified.py`, which drops samples exceeding `max_seq_length` (3,405 of 70,748 samples, 4.8%). Qwen CWM seeds use `train_repo_cwm.py`, which truncates and retains all samples. This difference affects total training steps (∼12,627 vs. ∼13,266 for DeepSeek, ∼6,633 for Qwen) but not the training data distribution, as dropped samples are uniformly distributed across labels.

Prompt templates. Prompt templates for all formulations (CLS, CWM, TRAJ) are provided in the code repository. CLS uses a binary classification template ("Does this test pass or fail?"); CWM uses an execution simulation template ("What is the output of this test?"); TRAJ uses a variant-level verdict template ("Is this patch resolved or unresolved?").

Mutation operators. We use five mutation types: (1) *gold*: unmodified code (all tests pass); (2) *operator_swap*: replaces comparison or arithmetic operators (e.g., < → <=);

Table 11: Training hyperparameters for CLS/CWM configurations. TRAJ models use lr $= 2\times10^{-4}$ due to the smaller training set (∼8K function-level instances vs. ∼70K for CLS/CWM); other hyperparameters are shared.

Shared across all configurations	
LoRA rank / alpha / dropout	16 / 32 / 0.05
LoRA targets	`q/k/v/o_proj`
Optimizer	AdamW (β_1=0.9, β_2=0.999)
LR scheduler	Linear
Effective batch size	32
Max sequence length	2,048
Epochs	3
Precision	bf16
Gradient checkpointing	✓
Seeds (function)	{42, 123, 456, 789, 0}[b]
Seeds (repo)	{42, 123, 456}[c]

Model-family–specific parameters			
	Qwen3-4B	**Qwen3-8B**	**DS-6.7B**
Learning rate	5e-5	5e-5	2e-4
Weight decay	0.01	0.01	0.0
Warmup ratio	0.05	0.05	0.1
Batch / grad. accum. (CLS)	2 / 16	1 / 32	2 / 16
Batch / grad. accum. (CWM)	1 / 32	1 / 32	1 / 32
Training samples	70,748	70,748	67,343–70,748[a]

[a]DeepSeek CLS canonical run (s42) uses drop preprocessing, which excludes 3,405 sequences exceeding max length (67,343 of 70,748). The s123 run uses truncate preprocessing (70,748 samples) but produces NaN validation loss. See §3.1. [b]Seeds 789 and 0 use NF4 quantization; others use bf16. [c]Repo-level seed coverage varies by configuration; see Table 3 caption for per-model counts.

(3) *off_by_one*: shifts loop bounds or array indices by ±1; (4) *return_value*: modifies return statements (e.g., negating a boolean, changing a constant); (5) *boundary_removal*: removes conditional guards or boundary checks. Mutation operators were applied using AST-level transformation; variants that did not alter any test outcome were discarded.

Compute budget. All experiments were conducted on NVIDIA RTX 5090 GPUs. Training time is approximately 1.5 hours per run for 4B models and 2 hours for 8B models. Total compute budget: ∼370 GPU-hours across 223 completed runs (135 successful, 88 failed/killed; ∼46 unique configurations including seeds, formulations, model sizes, and ablations).

F Prompt Templates

We show representative prompt examples for each formulation. All examples use the same input instance for comparability.

CLS (per-test classification).

```
System: You are a code verifier.
Given a code patch and a test name,
predict whether the test will pass
or fail when run against the
patched code.

User:
## Patch (git diff)
<patch content> [... truncated]
## Test
test_boundary_check
Will this test pass or fail?

Target: pass
```

CWM (execution simulation).

```
System: You are a code execution
simulator. Given a code snippet and
a test case, predict the execution
result.

User:
## Code
<code content> [... truncated]
## Test
test_boundary_check
What is the execution result?

Target: PASS: test completed
successfully
```

TRAJ (trajectory scoring).

```
System: You are a code repair
verifier. Given the code patch and
test summary, predict whether all
tests pass (resolved) or any test
fails (unresolved).

User:
[Code]
<patch content> [... truncated]
[Test Summary]
<all test names and descriptions>

Target: resolved
```

CLS and CWM share the same input format (one code–test pair per instance); CLS produces a single token (`pass`/`fail`), while CWM generates a variable-length execution trace. TRAJ receives all tests concatenated and produces a single token (`resolved`/`unresolved`).

G Val-Test Mutation Type Distribution

Table 12 reports the mutation type composition of the repo-level validation and test sets. The most notable shift is `operator_swap`, which nearly doubles from val (19.1%) to test (34.3%), while `off_by_one` shrinks from 10.0% to 1.9%.

The `operator_swap` shift is the primary source of val-test distribution mismatch. Since CLS achieves moderate accuracy on this type (78.9–81.4%), the over-representation in the test set does not systematically bias overall accuracy

Table 12: Repo-level mutation type distribution: validation vs. test set, with per-type CLS accuracy at two model sizes (seed 42). Shift = test% − val%.

Mutation Type	Val (%)	Test (%)	Shift	CLS 4B	CLS 8B
gold	45.2	47.4	+2.2	98.3	96.2
operator_swap	19.1	34.3	+15.2	78.9	81.4
boundary_removal	15.0	10.7	−4.3	46.9	46.9
return_value	10.7	5.7	−5.0	68.0	75.4
off_by_one	10.0	1.9	−8.1	58.5	86.9

estimates in either direction. Two boundary observations are notable: (1) `boundary_removal` represents a scale-invariant difficulty ceiling at ∼47% across both model sizes: this mutation type resists capacity scaling; (2) `off_by_one` shows the largest scaling gain (+28.4 pp, 58.5%→86.9%) but its small test share (1.9%) limits its contribution to overall accuracy differences across scales.

H Qualitative Error Analysis

To complement the quantitative divergence reported in §3.5, we examine representative errors from each formulation at repository level.

CWM failure modes. Two patterns account for the majority of unparseable CWM outputs on Qwen3. (1) *Task misunderstanding*: the model generates a complete simulated test runner log (test names, timing, summary lines) rather than a pass/fail verdict. For instance, on `django__django-14315`, the model outputs a fabricated 1,248-token Django test execution trace with hallucinated test names that do not exist in the target file. The model confuses "predict test outcome" with "simulate test execution environment." (2) *Autoregressive degeneration*: the model enters a repetitive loop (e.g., repeating `from _pytest.config import _parse_pytest_args` until truncation at 4,542 tokens). Both patterns produce zero usable signal and default to the majority-class label. This explains why unparseable predictions carry higher accuracy than parseable ones (§3.5).

CLS failure modes. CLS errors concentrate in two categories requiring multi-step reasoning. False positives arise from *cross-concern value propagation*: a boundary removal mutation in `pytest` converts a guarded attribute access (`if xfailed and not xfailed.run`) to `if True:`, causing a NoneType error on tests exercising unrelated skip marks. The model fails to trace the None path through the pytest hook lifecycle. False negatives arise from *mathematical equivalence*: on `HumanEval/130` (Tribonacci variant), the model predicts failure for correct code because verifying the self-referential recurrence requires algebraic substitution rather than syntactic inspection. This single problem accounts for 17% of all function-level false negatives.

Complementarity. These failure patterns operate at different levels: CWM fails at the format level (producing no usable signal), while CLS fails at the reasoning level (producing a confident but incorrect binary verdict). Both formulations are FP-dominant (over-predict failure), but CWM's higher FP rate (70.3% vs. 56.5%) means it over-predicts failure on instances CLS correctly classifies, producing partial complementarity (oracle combination: 90.63%; §I, paragraph a).

I Extended Analysis

Function-level CWM metric robustness. At function level, CWM outputs are short status strings (mean ∼5–15 tokens). The unparseable rate is only 1.39%, and default-to-pass bias is <1 pp—well below the SESOI threshold. Exact string matching is appropriate for these short, structured outputs; semantic-but-format-different matches are rare given the constrained output space.

(a) Error complementarity. CLS and CWM are both FP-dominant at function level but at different rates, producing partial complementarity. Among CLS errors (seed 42), 56.5% are false positives (PASS → FAIL); among CWM errors, 70.3% are false positives. Both formulations over-predict failure, but CWM's stronger FP bias means it incorrectly rejects passing tests that CLS correctly accepts. An oracle combination achieves 90.63% accuracy (1713/1890 correct, 177 both-wrong; +2.80 pp over the best single formulation, CLS 87.83%; single-seed estimate, seed 42). The complementarity arises from different *degrees* of fail-bias rather than opposite error directions: CWM's generation bottleneck produces aggressive failure predictions, while CLS's single-token output is more calibrated. Ensemble approaches exploiting this degree difference are a natural direction.

(b) CWM error mode breakdown. Among unparseable CWM outputs (Qwen3, repo-level), 76.9% contain no execution keywords; 22.9%

contain ambiguous co-occurring pass/fail keywords. Format compliance varies sharply across repositories (scikit-learn: near-zero unparseable; sphinx: 100%); repository-specific code patterns interact with the model's generation capabilities in complex ways. This per-repository variation is consistent with a model/tokenizer-specific failure rather than a formulation-level one.

(c) Relaxed matching does not rescue CWM. Relaxed matching (tolerating format deviations, L0–L4) *decreases* CWM accuracy: 58.04%→56.98% (seed 42) and 59.09%→55.91% (seed 123). Unparseable predictions default to the majority class (pass, 72.96%), which outperforms the model's own parseable predictions (38–43%). The divergence widens from 26 pp to 27 pp under relaxed matching. This rules out the hypothesis that parsing strictness inflates the observed gap. Constrained decoding (Willard and Louf, 2023) could enforce format compliance but would not address the semantic accuracy deficit among parseable outputs. An accidental training artifact provides further evidence: a CWM model inadvertently trained on function-level (CLS-format) data achieved 86.05% accuracy with 0% unparseable outputs at repo level—recovering CLS-level performance. This confirms that the repo-level CWM degradation stems entirely from trace generation complexity, not from the underlying verification task.

(d) Variance compression. CWM's 5-seed standard deviation (0.37 pp) is lower than CLS (0.78 pp) and TRAJ (0.97 pp) at function level. If CWM performance is capped by output-format constraints at a given complexity level, seed-to-seed variability is compressed relative to formulations whose performance depends more on learned representations.

(e) Per-test precision and recall. CLS achieves fail F_1 87.59% (precision 86.20%, recall 89.04%) at seed 42.Multi-F2P instances (6–10 tests) reach 96.3% accuracy, consistent with the oracle finding that per-test granularity is most valuable for complex instances.

(f) Block correlation. Within-problem errors are correlated (ICC = 0.191, DEFF = 3.97, 2.81× more all-correct problems than i.i.d. expectation). All statistical tests use clustered bootstrap accordingly.

(g) Failed interventions. We systematically tested four interventions targeting different hypothesized causes of CWM repo-level degradation; all failed:

- **Capacity (rank 32, α=64):** −0.47 pp, within seed variance (§B).
- **Majority voting (K=3):** −0.68 pp; 93.3% unanimous, net flips negative (§B).
- **Relaxed matching (L0–L4):** −1.06 pp; default-pass outperforms parseable predictions (paragraph (c) above).
- **Model scaling (4B→8B):** parseable accuracy decreases (42.88%→38.52%) while unparseable rate rises (53.7%→73.8%); gain is a default-assignment artifact (§3.5).

Each intervention targets a distinct alternative explanation (insufficient capacity, random noise, parsing strictness, model scale); the uniform failure across all four confirms the bottleneck is formulation-level—output space complexity at repo level exceeds what ≤8B generation models can handle, independent of these surface-level factors.

(h) Per-mutation scaling patterns. Scaling benefits are highly non-uniform across mutation types (Table 8). `off_by_one` shows the largest CLS scaling gain: 58.5% (4B) to 86.9% (8B, +28.4 pp). In contrast, `boundary_removal` is scale-invariant at ∼47% across both sizes, which marks an intrinsic difficulty ceiling. For CWM, gold mutation parse rate collapses from 43.5% (4B) to 23.9% (8B) despite being the simplest target: generation capability degrades even for trivially correct code at larger scale within Qwen3. The val-test mutation type distribution differs notably (`operator_swap` shifts from 19.1% to 34.3%; Table 12), but this shift does not systematically bias overall accuracy because CLS performance on `operator_swap` (78.9–81.4%) is close to the overall mean.

(i) Inference efficiency. CLS achieves 217× higher throughput than CWM at repo level (853 vs. 3.9 inferences/min; Qwen3-4B, RTX 5090, 100 samples, avg ∼1,040 input tokens). The speedup derives directly from the formulation: classification requires a single output token, whereas CWM generates an average of 477 tokens per inference. Median latency is 48ms (CLS) vs. 16.3s (CWM). At the observed unparseable rate of 73.8% (8B),

the majority of CWM's generation budget produces tokens that cannot yield a valid verdict. This CLS–CWM efficiency gap is an inherent property of the formulation difference (1 vs. $O(L)$ output tokens) and should generalize across models, unlike the accuracy divergence which may be model-contingent.

The CLS–TRAJ efficiency comparison is more nuanced, because CLS makes one call per test while TRAJ makes one call per instance. Per-call, CLS is 20–138× faster (depending on input length: 47ms for short contexts vs. 2,632ms for TRAJ). However, the per-instance effective speedup depends on test count $|T|$: at function level ($|T| \approx 2.3$), CLS is ∼24× faster (108ms vs. 2,632ms per instance); at repo level ($|T| \approx 12.3$, ∼70ms/call with longer inputs), CLS is ∼3× faster (861ms vs. 2,631ms). A crossover exists at $|T| \approx 50$ tests, beyond which TRAJ's single-call amortization dominates. This boundary condition is informative for deployment: CLS is strongly preferred for function-level and moderate repo-level test suites, while extremely large test suites (>50 tests per instance) favor trajectory-level efficiency despite its coarser diagnostic signal.

(j) Re-repair pilot. We tested whether per-test diagnostic feedback improves downstream repair. A Qwen3-8B repair agent received either per-test diagnostic signals (which specific tests failed and why) or binary accept/reject feedback, evaluated on 44 SWE-bench Verified instances. The diagnostic condition achieved accuracy no higher than the binary baseline ($A_1 = 52.3\%$, $p = 0.42$, effect size ≈ 0). Per-test diagnostic information does not automatically translate to repair improvements at this scale. The negative result may reflect insufficient agent capability to exploit fine-grained feedback, insufficient sample size (at $n = 44$, statistical power is ∼0.3 for a medium effect size), or a genuine limitation of the diagnostic-to-repair translation.

(k) Relationship to chain-of-thought verification. CWM is structurally a chain-of-thought verifier: the model generates an execution trace token by token and derives its PASS/FAIL verdict from the final trace state. If intermediate reasoning improved verification, CWM should outperform the reasoning-free CLS baseline. At function level, no such advantage materialises: CWM and CLS are statistically equivalent (TOST p=0.007, ±2 pp margin; §3.4), indicating that the additional trace generation provides no accuracy benefit over direct binary classification. At repo level, the chain-of-thought mechanism actively *hurts*: trace generation triggers format collapse (53–74% unparseable), and even parseable traces yield accuracy below the majority baseline (38–43% vs. 72.96%; paragraph (c)). Four targeted interventions fail to rescue the generation pathway (paragraph (g)). Within the ≤8B LoRA regime, chain-of-thought–style verification does not outperform direct classification for execution prediction. Whether execution-trace pretraining at larger scale (FAIR CodeGen team et al., 2025) changes this conclusion remains the primary open question (§3.7).

J TOST Margin Sensitivity

Table 13 reports the TOST equivalence p-value as a function of the equivalence margin, computed on the 5-seed variant-level CLS–CWM comparison (gap = 1.24 pp, 523 variants, 114 clusters, clustered bootstrap).

Table 13: TOST margin sensitivity for CLS–CWM variant-level equivalence (5 seeds, 523 variants, 114 clusters). The smallest margin achieving equivalence at $\alpha = 0.05$ is ±1.52 pp.

Margin (± pp)	p-value	Equivalent (α=0.05)?
1.0	≫0.05	No
1.5	0.054	No (borderline)
1.52	**∼0.05**	**Smallest margin**
1.71	0.027	Yes
2.0	0.007	Yes
3.0	4.6×10^{-5}	Yes
5.0	≪0.001	Yes

The minimum equivalence margin is ±1.52 pp, 0.48 pp below our pre-specified ±2 pp threshold. The ±2 pp threshold was set based on deployment relevance rather than observed data: at function-level accuracy (85–88%), a 2 pp per-test difference corresponds to ∼38 additional errors across 1,890 tests (∼2–3 problems out of 114), comparable in magnitude to between-seed standard deviation (0.37–0.97 pp) and checkpoint selection noise (<1 pp). Differences within this range do not reliably distinguish formulations from training noise in pipeline selection decisions. For CLS–TRAJ, the TOST at ±2 pp yields p=0.025, nominally significant but not surviving Holm-Bonferroni correction at k=5 (adjusted p=0.075; Table 15).

Robustness: seed-level vs. clustered bootstrap TOST. The primary TOST p-values use seed-level paired analysis (df=4), treating each training seed as the unit of replication (§3.1). Under problem-level clustered bootstrap (resampling 114 problem clusters, 10K iterations), the CLS–TRAJ raw TOST p decreases to $\sim$0.007 (Holm-adjusted $p\approx$0.021), which would confirm equivalence. We report the more conservative seed-level result (p=0.025, adjusted p=0.075) as primary, since training seeds are the true unit of replication; problem-level bootstrap pools within-seed observations and may understate between-seed uncertainty. For CLS–CWM, both methods yield p=0.007, so the choice does not affect conclusions.

J.1 Bayesian ROPE Analysis

To complement the frequentist TOST, we conduct a Bayesian equivalence analysis using a paired t-test with Jeffreys prior (Kruschke, 2013, 2018). The marginal posterior for the mean accuracy difference follows a t-distribution with $n-1$ degrees of freedom. For CLS–CWM, the posterior probability within the $\pm$2 pp ROPE is $P(\text{ROPE}) = 0.975$, and the 95% highest density interval (HDI $= [+0.10, +2.00]$ pp) falls entirely inside the ROPE, confirming equivalence under both continuous and HDI-based decision criteria. For CLS–TRAJ at per-test level, $P(\text{ROPE}) = 0.542$ (inconclusive; 95% HDI $= [+0.35, +3.52]$ pp extends beyond the ROPE), consistent with the TOST non-confirmation after Holm correction ($p_{\text{adj}} = 0.075$). At variant level, CLS–TRAJ equivalence is supported by the Bayesian criterion ($P(\text{ROPE}) = 0.970$, HDI $= [-2.00, +1.03]$ pp), but the frequentist TOST does not confirm equivalence after Holm-Bonferroni correction ($p_{\text{adj}} = 0.075$; Table 15). This discrepancy reflects differing treatments of multiplicity: Bayesian ROPE evaluates each comparison independently, while Holm correction penalizes across all k=5 primary tests. For CLS–CWM, both frameworks agree (TOST p_{adj}=0.028, $P(\text{ROPE})$=0.975); for CLS–TRAJ per-test, both are inconclusive. We conservatively follow the Holm-corrected TOST in the main text and report CLS–TRAJ equivalence as not confirmed.

14B scaling evidence. At 14B (3 seeds), the CLS–TRAJ gap narrows to 1.07 pp (95% CI $[-0.97, +2.19]$ pp). Bayesian ROPE gives $P(\text{ROPE})$=0.956, and raw TOST confirms equivalence (p=0.043). This is a separate correction family (k=4) covering the four function-level equivalence tests across 8B and 14B scales (CLS–CWM and CLS–TRAJ at each scale); Holm correction yields adj p=0.050, at the conventional boundary. The monotonic gap reduction (3.11 pp at 4B $\rightarrow$ 2.24 pp at 8B $\rightarrow$ 1.07 pp at 14B) and consistently high $P(\text{ROPE})$ at 14B favor a capacity-dependent convergence interpretation, though the Holm-corrected result remains borderline.

K Variant-Level Effect Sizes

Table 14 reports variant-level bootstrap confidence intervals for the repo-level formulation comparisons (n=48 variants). Following Hoenig and Heisey (Hoenig and Heisey, 2001), we report minimum detectable effect (MDE) rather than post-hoc observed power.

Table 14: Variant-level bootstrap analysis (n=48). CLS–CWM uses paired bootstrap (same variants, same seed); CLS–TRAJ uses unpaired bootstrap (per-variant TRAJ predictions not archived for pairing). MDE at n=48: $\sim$25 pp.

Comparison	Config	Δ (pp)	95% CI	p
CLS *vs.* CWM *(paired)*				
	Qwen3-8B	+29.17	[10.42, 47.92]	0.003
	DeepSeek-6.7B	+35.42	[18.75, 50.05]	<0.001
CLS *vs.* TRAJ *(unpaired)*				
	Qwen3-4B	−4.17	wide	0.740
	Qwen3-8B	+8.33	wide	0.385
	DeepSeek-6.7B	0.00	wide	1.000

At n=48, the MDE is $\sim$25 pp. The CLS–CWM gap (29–35 pp) exceeds this threshold, confirming that n=48 is sufficient to detect the observed divergence. The CLS–TRAJ gaps (<10 pp) fall well below the MDE: any true difference is too small to detect at this sample size, consistent with the function-level TOST equivalence (§3.4).

The CLS–TRAJ unpaired bootstrap produces wider confidence intervals than paired analysis would; TRAJ per-variant predictions were not archived in a format that permits pairing. The CLS–CWM paired comparison uses CLS s123 vs. CWM s42 (different seeds), but the CI lower bound of 10.42 pp far exceeds the seed noise observed in function-level experiments (<2 pp).

L Multiple Comparison Correction

Table 15 applies Holm-Bonferroni correction (Holm, 1979) to the five primary statistical

comparisons across this study (k=5, family-wise α=0.05).

Table 15: Holm-Bonferroni correction for primary comparisons (k=5). Comparisons ranked by raw p-value; each tested against $\alpha/(k - \text{rank} + 1)$.

Rank	Comparison	Raw p	Factor	Adj p	Sig.?
1	Repo: CLS vs CWM	0.003	×5	0.015	✓
2	Func: CLS vs CWM (TOST ±2 pp)	0.007	×4	0.028	✓
3	Func: CLS vs TRAJ (TOST ±2 pp)	0.025	×3	0.075	—
4	Repo: CWM vs TRAJ	0.039	×2	0.078	—
5	Repo: CLS vs TRAJ	0.385	×1	0.385	—

The two core claims survive correction: the repo-level CLS–CWM divergence (adjusted p=0.015) and function-level CLS–CWM equivalence (adjusted p=0.028). The function-level CLS–TRAJ equivalence does not survive correction (adjusted p=0.075). Under problem-level clustered bootstrap, the raw TOST p decreases to ∼0.007 (adjusted p≈0.021), which would confirm equivalence; we report the more conservative seed-level result (p=0.025) as primary, since training seeds are the true unit of replication.

M Mixed-Effects Regression

We fit mixed-effects linear probability models (LPM) with heteroskedasticity-robust standard errors (HC1) to test the formulation×complexity interaction formally (Moulton, 1990). We choose LPM because coefficients are directly interpretable as probability differences (percentage points), consistent with the reporting units in the main text; at these sample sizes (n≈56K cross-complexity, repo-level pooled cross-model) and base rate range (0.6–0.9), LPM and logistic regression yield concordant conclusions for all primary effects (Angrist and Pischke, 2009) (confirmed below).

Cross-complexity model (DeepSeek-Coder). `correct ~ formulation × complexity + (1|problem_id)`, n=56,480 observations (CLS + CWM × function + repo), 139 problem clusters. The CWM main effect at function level is non-significant ($\hat{\beta}$= − 0.002, z= − 0.25, p=0.805); the CWM×repo interaction is highly significant ($\hat{\beta}$= − 0.219, z= − 23.7, p<0.001). CWM and CLS are statistically indistinguishable at function level (Δ= − 0.2 pp, p=0.805) but diverge by 22 pp at repo level, confirming the complexity-dependent divergence is not an artifact of aggregation.

Repo-level cross-model model. `correct ~ formulation × model + (1|problem_id)`, both model families, repo level only. CWM deficit: $\hat{\beta}$= − 0.22 (−22 pp, z= − 64.4, p<0.001). Model effect: +1.1 pp for Qwen3 (z=3.1, p=0.002 in LPM; not significant under GEE, p=0.449). Formulation×model interaction: +1.6 pp less CWM degradation on Qwen3 (z=2.8, p=0.006 in LPM; not significant under GEE, p=0.860). Formulation dominates the predictable variance; model identity is directionally positive but specification-dependent. The 25-cluster random intercept structure limits effective degrees of freedom for the random effects; inferences on fixed effects (formulation, complexity) are robust as they pool across the full observation count.

Logistic regression robustness check. To verify that LPM conclusions are not artifacts of the linear link function, we fit GEE logistic regression (binomial family, exchangeable working correlation, robust sandwich standard errors) to the same data. For the cross-complexity model, the CWM main effect at function level has OR=0.98 (95% CI [0.86, 1.12], p=0.771) and the CWM×repo interaction has OR=0.31 ([0.15, 0.65], p=0.002). Average marginal effects are −0.2 pp (function) and −21.6 pp (repo), matching LPM coefficients within 0.3 pp. For the cross-model model, the CWM deficit has OR=0.30 ([0.15, 0.63], p=0.001, AME= −21.1 pp). Direction, magnitude, and significance of all primary effects are concordant between LPM and logistic regression; minor discrepancies in secondary effects (model identity, formulation×model interaction: p=0.449 and p=0.860 under GEE vs. p=0.002 and p=0.006 under LPM) reflect the GEE's more conservative cluster-robust inference with k=25 clusters. The discrepancy in secondary effects reflects cluster count: the cross-complexity model (k=139) yields reliable sandwich SEs, while the cross-model model (k=25) falls below the k≥40 threshold recommended for stable cluster-robust inference (Cameron et al., 2008).

N CWM Parseable Bias Analysis

The default-to-pass assignment for unparseable CWM outputs is not random with respect to instance difficulty. Unparseable instances have significantly higher pass rates than parseable ones (74.6% vs. 68.3%, χ^2 $p = 5.9 \times 10^{-10}$), and CLS

models achieve higher accuracy on unparseable instances (86.4% vs. 79.1%, $p = 3.3 \times 10^{-18}$). This indicates that CWM's parsing failures concentrate on simpler, pass-heavy instances where the majority-class default happens to be correct more often. Cross-model Jaccard overlap of unparseable instance sets is 0.46, confirming this pattern is instance-driven (specific inputs consistently cause parsing failure across models) rather than stochastic. The net effect is that CWM's reported accuracy is inflated by default-correct assignments on easy instances, while its effective discriminative accuracy on harder instances is substantially lower than the aggregate figure suggests.

O Threats to Validity (Full Analysis)

We identify five potential confounds and assess their impact on our conclusions.

Checkpoint selection. Val-based selection is unavailable for runs with NaN validation loss (DeepSeek CWM repo-level runs, DeepSeek CLS s123), which default to the final epoch checkpoint. A checkpoint ablation on Qwen3-8B CLS s42 quantifies this confound: epoch 1 (ckpt-2200) achieves 85.34%, val-selected (ckpt-3600) 84.48%, and final epoch (ckpt-6633) 85.29%—a $<$1 pp range. The val-selected checkpoint is the lowest, indicating no upward bias from checkpoint selection. The Qwen3-8B TRAJ comparison (epoch 2: 72.92% vs. epoch 3: 70.83%) shows a slightly larger $\sim$2 pp shift. For CWM, the convergence analysis (Appendix O) shows eval loss at the final epoch is 1.8$\times$ its minimum; however, early-stopped checkpoints retain high unparseable rates, leaving the qualitative ranking unchanged.

CWM convergence analysis. Qwen3-8B CWM repo-level eval loss reaches its minimum within $\sim$5% of training (step 300–400 of 6,633) and increases monotonically thereafter, with no plateau or recovery (Table 16). This pattern is consistent across all three seeds, confirming systematic overfitting to trace generation rather than stochastic variation. The final-epoch eval loss is $\sim$1.8$\times$ the optimal value. By contrast, function-level CWM achieves CLS-equivalent accuracy (TOST p=0.007) with the same training recipe, indicating that the overfit is specific to repo-level output complexity. Cross-model validation: a DeepSeek-Coder early-stopped checkpoint (s123, epoch 1) achieves 75.45% versus 57.31% at epoch 3 (s123; +18.14 pp), but retains 66.4% unparseable rate, confirming that format collapse emerges early and is not a late-training artifact. No epoch-level checkpoint is available for Qwen3 CWM (only step-level checkpoints at steps 300/400 and late steps $\geq$5800; continuous epoch checkpointing was not enabled for this training run). The convergence analysis (Table 16) shows eval loss minimum at step 300–400 across all three Qwen3 seeds, consistent with DeepSeek-Coder's epoch-1 improvement, providing cross-model evidence that the overtraining pattern generalizes. An early checkpoint (step 400, $\sim$6.5% of training, s42) yields 65.17% repo-level accuracy with 63.7% unparseable rate—comparable to or below final-epoch performance (66.93–67.33% accuracy, 60.4–63.9% unparseable; s123/s456 at step 6633). This rules out late-training format degradation: collapse is present from early training, and the optimal eval-loss checkpoint does not recover repo-level accuracy.

Table 16: CWM repo-level eval loss across training (Qwen3-8B, 3 seeds). Minimum is reached at step 300–400 ($\sim$5% of training); subsequent training monotonically increases loss.

Step	s42	s123	s456
300	0.291	**0.288**	**0.277**
400	**0.286**	0.306	0.283
1000	0.333	0.336	0.351
3000	0.445	0.453	0.435
6600	0.521	0.517	0.507

Evaluation sequence length. CWM accuracy varies by $\sim$9 pp across evaluation sequence lengths for the same checkpoint (seq2048: $\sim$65% vs. seq4096: $\sim$56%), driven by unparseable-rate differences that propagate through majority-class default assignment. CLS is invariant to this parameter. Cross-model CWM comparisons use different eval lengths (Qwen3: 8192, DeepSeek: 4096) and are not directly comparable in absolute terms. A matched-sequence evaluation shows $\sim$1.5 pp difference for CLS (Appendix B), insufficient to explain the 18–26 pp gap. Furthermore, increasing Qwen3-8B CWM evaluation from seq4096 to seq8192 *raises* the unparseable rate (63.7%$\rightarrow$73.8%) while changing accuracy by only 0.81 pp (64.36%$\rightarrow$65.17%); giving the model more generation budget worsens format compliance rather than improving it, confirming that format collapse is a structural bottleneck

rather than a truncation artifact. Practitioners must standardize evaluation sequence length for generation-based formulations.

Preprocessing. DeepSeek CLS accuracy ranges from 82.61% (s42, drop preprocessing) to 74.93% (s123, truncate preprocessing), a 7.68 pp spread that conflates preprocessing strategy, checkpoint selection method, and seed. The two strategies produce different effective training sets (67,343 vs. 70,748 samples) and different checkpoint selection regimes. The qualitative ranking CLS > CWM holds under both strategies (Table 4).

TRAJ training regime. TRAJ uses a different training configuration: smaller training set (∼400 repo-level variants vs. ∼70K per-test instances), higher learning rate (2×10^{-4} vs. 5×10^{-5}), and variant-level labels. This asymmetry is inherent to the formulation (trajectory-level labels are only available at variant granularity), but precludes training-data-controlled comparison at repo level. The downsample ablation confirms the CLS formulation advantage persists at matched data volume (80.74% vs. 72.92%; Appendix B), though it controls for training volume but not label granularity (per-test vs. variant-level). Crucially, TRAJ achieves statistical equivalence with CLS at function level ($\leq$1.3 pp, §3.2), demonstrating the formulation itself is viable; the repo-level gap attribution remains uncertain: data volume (∼400 variants vs. ∼70K per-test instances) and label granularity (variant-level vs. per-test) are both plausible confounds that cannot be separately controlled.

CWM scale and pretraining. Our CWM baseline uses 4–8B LoRA fine-tuning without execution trace pretraining. The original Code World Model (FAIR CodeGen team et al., 2025) uses 32B parameters with mid-training on execution traces, a fundamentally different training regime. At larger scale or with execution-trace pretraining, the generation bottleneck observed here may diminish: larger models produce more coherent long-form outputs, and execution trace exposure may improve output format compliance. This is the primary limitation: all repo-level CWM conclusions are conditional on the $\leq$8B LoRA regime.

Despite these confounds, the repo-level CLS–CWM gap (18–26 pp) exceeds any single assessed confound. Even under worst-case compounding (∼11.5 pp), the gap retains $\geq$7.5 pp margin; confounds predominantly disadvantage CWM. However, correlated confounds and limited seed coverage mean definitive causal attribution requires multi-seed replication. At n=48, the minimum detectable effect (MDE) is ∼25 pp (Hoenig and Heisey, 2001); the CLS–CWM gap (29–35 pp) exceeds this threshold, while the CLS–TRAJ gap ($<$10 pp) falls below it, consistent with our TOST equivalence finding. Holm-Bonferroni correction across the five primary comparisons (k=5; Table 15) confirms the CLS–CWM gap (adjusted p=0.015) and function-level CLS–CWM equivalence (adjusted p=0.028); function-level CLS–TRAJ equivalence does not survive correction (adjusted p=0.075), nor does the repo-level CWM–TRAJ comparison (adjusted p=0.078). The consistent direction across both model families and all tested configurations supports a formulation-level explanation, though definitive causal attribution requires future experiments addressing each confound individually.

Extended limitations. The same-size 4B comparison uses a single seed; CLS 4B seed 123 (val-best ckpt-800) achieves only 75.33% test accuracy (+2.4 pp above majority baseline); val-best selection at an early checkpoint (step 800) amplifies the val-test distribution shift observed across all configurations. The CWM 8B comparison uses 3 seeds per model. The validation and test sets have zero problem overlap but differ in mutation type composition (Table 12): `operator_swap` shifts from 19.1% (val) to 34.3% (test), the largest single-type distribution mismatch (+15.2 pp). All repo-level results use val-selected checkpoints, and checkpoint selection variance could contribute to observed differences. Our training data is synthetically constructed from mutation operators rather than real developer patches. While mutation testing literature establishes that mutants are reasonable proxies for real faults in test effectiveness studies (Andrews et al., 2005; Just et al., 2014), these findings concern mutation *adequacy* (whether tests that kill mutants also detect real faults), not whether verifiers trained on mutants transfer to real patches. However, the coupling effect underlying these results supports *relative ranking* validity: a domain shift from synthetic mutations to real bugs may affect absolute accuracy levels, but our core finding—that formulation choice matters at repo level but not function level—concerns relative performance ordering, which is expected to be preserved un-

der distributional shifts that affect all formulations equally. The synthetic-to-real transfer gap for absolute accuracy remains untested. The TRAJ training configuration differs from CLS/CWM: smaller training set (∼400 vs. ∼70K instances), different learning rate (2×10^{-4} vs. 5×10^{-5}), and variant-level evaluation (n=48 vs. per-test n=9,784). The function-level CLS–TRAJ TOST equivalence ($p = 0.025$, seed-level paired, ±2 pp) is computed on 523 matched variant-level instances across 5 seeds; 2 of 5 seeds use NF4 quantization. Restricting to three bfloat16 seeds yields a comparable mean gap (-0.45 pp) but wider confidence interval (df=2); the TOST p-value increases to 0.128, which reflects reduced statistical power rather than a shift in the point estimate. The NF4 seeds contribute precision (narrowing the CI) without shifting the mean, consistent with precision being comparable across both quantization levels at function level. Repo-level CLS–TRAJ comparison should be interpreted with caution due to configuration and granularity differences.